THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE TWO-CHAPTER TEST SET A

Chapters 7 & 8: **How do Organisms Reproduce?, Heredity** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which organism reproduces by regeneration? 1

(a) yeast **(b)** Planaria **(c)** Amoeba **(d)** Rhizopus

2. The part of the flower that develops into a fruit is the: 1

(a) ovary **(b)** anther **(c)** ovule **(d)** stigma

3. Pollen grains are produced in the: 1

(a) ovary **(b)** style **(c)** stigma **(d)** anther

4. A student states: “Hydra reproduces by budding as well as regeneration.” The correct response is: 1

(a) Correct **(b)** Incorrect

(c) Only budding **(d)** Cannot be determined

5. The method of vegetative propagation in sugarcane is: 1

(a) seeds **(b)** stem cutting **(c)** leaf buds **(d)** roots

6. The physical expression of a gene is the: 1

(a) allele **(b)** chromosome **(c)** phenotype **(d)** genotype

7. The ratio of tall to short plants in the F2 of a monohybrid cross is: 1

(a) 1 : 2 : 1 **(b)** 9 : 3 : 3 : 1 **(c)** 1 : 1 **(d)** 3 : 1

8. The term for a pair of contrasting forms of a gene is: 1

(a) zygotes **(b)** genomes **(c)** chromosomes **(d)** alleles

9. A student states: “Each human gamete contains one sex chromosome.” The correct response is: 1

(a) Correct **(b)** Only in males

(c) Incorrect **(d)** Cannot be determined

10. The probability that a child of XX × XY parents will be female is: 1

(a) 1/2 **(b)** 1/4 **(c)** 1 **(d)** 3/4

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What is the difference between a gamete and a zygote? 2

12. Name two plants propagated by grafting or layering. 2

13. What is a Punnett square? 2

14. What is the F1 phenotype when pure tall (TT) is crossed with pure short (tt)? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Why is sexual reproduction more beneficial than asexual? Give three reasons. 3

16. Explain with an example how genes control the characteristics (traits). 3

17. What are the functions of the 3
(i) fallopian tube
(ii) uterus
(iii) placenta?

OR

What is budding? Explain with Hydra and a diagram. 3

18. In a cross between Tt and Tt, find the genotypic and phenotypic ratios. 3

OR

What are the differences between inherited traits and acquired traits? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Explain the structure of a flower with a diagram. 5
(ii) Distinguish between self- and cross-pollination.
(iii) What is a unisexual flower?

OR

Case study: A gardener grows rose plants by cutting stems and also by layering. 5
(i) Name the type of reproduction.
(ii) Give two advantages.
(iii) Define layering.
(iv) Why can't bananas be grown from seeds?
(v) Name one more plant grown this way.

20. (i) Describe Mendel's experiment with tall and short plants. 5
(ii) Write the F1 and F2 phenotypes and genotypes.
(iii) What did he conclude?

OR

Case study: A geneticist crosses a pea plant with round yellow seeds (RrYy) with another RrYy plant. 5
(i) Write the gametes.
(ii) What is the F2 phenotypic ratio?
(iii) How many are wrinkled green out of 16?
(iv) What law does it illustrate?
(v) What proportion is round?

END OF QUESTION PAPER — Total: 40 Marks